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Roll no.

SUMMER-SEMESTER, B. Tech.

END TERM EXAMINATION, July - 2026

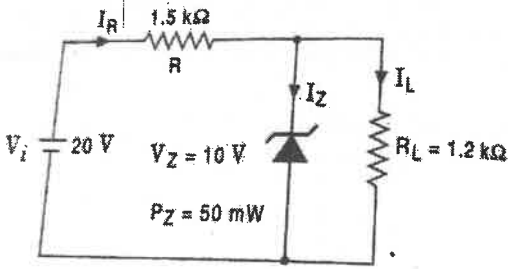
Paper Code – ECECC:202/EIECC:202/ICECC:202/VTECC:202

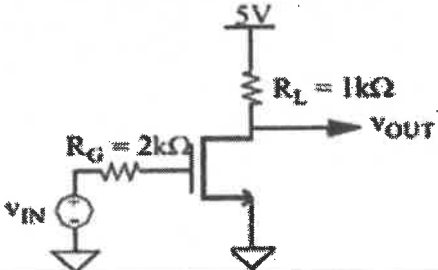
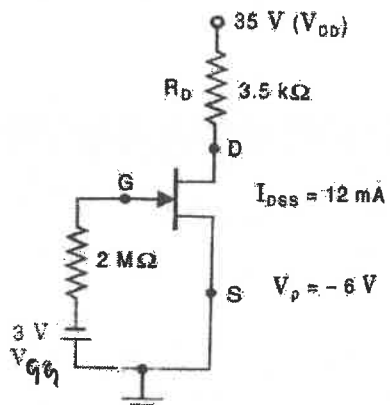
Subject: **Electronic Devices & Circuits**

Time: 3 Hours

Max. Marks - 50

Note: - Missing data/ information if any, may be suitably assumed & mentioned in the answer. All questions carry equal marks [10=5+5].

	There are three parts in each question. Attempt any two parts.	CO
1a	Explain the construction and working of a p-n junction diode. Derive the Shockley diode equation and explain the V-I characteristics of the diode under forward and reverse bias conditions.	CO1
1b	The electron mobility and hole mobility of silicon are: $\mu_n = 1350 \text{ cm}^2/\text{Vs}$, $\mu_p = 480 \text{ cm}^2/\text{Vs}$ given ($n = 5 \times 10^{15} \text{ cm}^{-3}$, $p = 2 \times 10^{15} \text{ cm}^{-3}$, $q = 1.6 \times 10^{-19} \text{ C}$) Calculate the conductivity of the semiconductor.	CO1
1c	An intrinsic silicon (Si) crystal is doped with donor impurities to a concentration of $1.4 \times 10^{24} / \text{m}^3$ when the intrinsic carrier concentration is $1.4 \times 10^{18} / \text{m}^3$. Find the equilibrium concentration of holes (p) and ratio of the electron to hole concentration.	CO1
2a	A 100 V, 50 Hz AC supply is connected to a bridge rectifier with a load resistance of $2 \text{ k}\Omega$ and transformer turn ratio 10:1, Calculate DC Load Current (I_{dc}), rms current (I_{rms}), ripple factor and efficiency of the bridge rectifier. Assume ideal diodes.	CO2
2b	For a given Zener diode network determine V_L , V_R , I_Z and I_R . 	CO2
2c	Explain the concepts of drift current and diffusion current in semiconductors. Also, sketch and describe the V-I characteristics of an ideal diode, silicon diode, LED, and Zener diode.	CO2
3a	Derive the expression for the drain current in a MOSFET. Also, sketch the V-I characteristics and clearly indicate the different regions of operation.	CO3

3b	Find the small signal parameters g_m and dc voltage V_{GS} and V_{out} of this amplifier with $\mu_n C_{ox} = 100 \mu A/V^2$, $(W/L) = 6/0.1$, $V_t = 0.5 V$, $I_D = 0.275 mA$.	CO3
		
3c	Determine the following parameter for a given network (a) V_{GSQ} (b) V_{DS} (c) V_D (d) V_G (e) V_S	CO3
		
4a	Explain the input and output characteristics of a Bipolar Junction Transistor (BJT) operating in Common Emitter (CE), also find the equation for input resistance, output resistance, current gain.	CO4
4b	Determine I_E , α and β of common-base transistor circuit (Given: $I_C = 7mA$, $I_B = 0.1mA$).	CO4
4c	The collector and base current of an N-P-N transistor are measured as $I_C = 5 mA$, $I_B = 50 \mu a$, and $I_{CBO} = 1 \mu a$. a. Determine α , β and I_E b. Determine the new level of I_B required to produce $I_C = 10 mA$.	CO4
5a	Explain the concept of virtual ground and CMMR in detail.	CO5
5b	Draw the circuit diagram of adder and subtractor and find out the expression for output.	CO5
5c	(i) The input voltage to the circuit is given by $v_i = 0.1 \sin 2000\pi t$ V. What is the output voltage if $R = 10 k\Omega$, $C = 0.005 \mu F$, and $v_o(t=0) = 0$? (ii) What is the voltage transfer $V_o(s)/V_i(s)$ function?	CO5
	